RAG pipeline for the generation of ER discharge recommendations

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Abstract

Emergency Room discharge recommendations are often written under severe time pressure, resulting in documents that may lack clarity, completeness, and patient accessibility. This work introduces a retrieval-augmented generation (RAG) pipeline designed to automatically produce patient-friendly discharge recommendations grounded in clinical guidelines. The system retrieves relevant excerpts from a corpus of official European medical guidelines and uses a Large Language Model (LLM) to generate structured, actionable recommendations supported by traceable references.
Multiple LLMs have been used in testing, including smaller local models and large API-based models, with high-end models providing clean outputs with minimal editing needs.

Limitations arise from the guideline corpus, including incomplete coverage and the loss of structured information during image and table parsing, as well as from the small evaluation set and hardware constraints that limit local deployment. Despite these challenges, the system shows strong promise: top-performing models generate clinically coherent and readable recommendations, nearly ready for deployment, within timeframes compatible with real-world workflows.
Overall, the pipeline demonstrates solid potential as a clinical decision support tool.

1 Introduction

In Italian hospitals, patients discharged from the Emergency Room (ER) receive a document summarising their visit. This document includes the reason for admission, the patient's medical history, the objective examination findings, the clinical diary documenting each step of the care process, a diagnosis, a prognosis, and a set of discharge recommendations. This document is addressed to both the patient and their general practitioner responsible for follow-up care, and is the primary communication medium for outpatient management's instructions.
Due to the frenetic nature of Emergency Rooms, physicians often lack the time to draft thorough and clearly worded discharge recommendations. Some recommendations are only communicated orally and risk being forgotten by the patient, particularly when they are elderly, distressed, or otherwise impaired. Other times, especially for less important instructions, they are only implied by the physicians and never communicated to the patient. Recommendations that are written on the document tend to contain specific medical abbreviations, which might be hard to interpret for the common citizen, may lack sentence coherence and/or omit relevant secondary guidance such as lifestyle modifications.
This is not due to negligence on the physician's part, but a direct consequence of the time constraints inherent to an ER department. Nonetheless, it may still cause problems to the patient and reflect poorly on the attending physician and the medical institution itself.

The emergence of Large Language Models (LLMs) has introduced new possibilities for clinical text generation, and is already aiding many doctors in producing more readable documentation. However, commercial general-purpose models present significant limitations in this context. Their tendency to hallucinate clinical facts, combined with the opacity of their training data, makes them unsuitable for producing medical recommendations that physicians can trust and sign. Even their "Search the Web" function, a more transparent generation feature that provides links to the sources, often fishes from untrusted domains, with their most cited source being Reddit(analyticsinsight). Consequently, while

LLMs possess the linguistic capability to assist in this task, their application in a clinical setting demands a more controlled, source-grounded approach.

This project proposes a Retrieval-Augmented Generation (RAG) pipeline designed to address this gap. The system takes as input structured clinical information that physicians already produce during the ER visit (medical history, objective examination, clinical diary, diagnosis, and prognosis) and retrieves relevant passages from a curated corpus of official European medical guidelines. An LLM then generates clearly written discharge recommendations with information retrieved exclusively from the guideline excerpts, ideally covering both pharmacological therapy and lifestyle factors, including those that are often omitted under time pressure. Each recommendation is accompanied by a traceable source reference and a brief clinical justification, enabling the reviewing physician to verify the rationale behind every suggestion before delivering the document to the patient.
The physician using this system would then only have to check the accuracy of the recommendations before attaching them to the official document, saving time while simultaneously improving the quality and readability of the produced recommendations.

2 Theoretical Framework

This section gives a brief introduction of the core concepts and techniques on which the system's architecture is based.

- **Retrieval-Augmented Generation (RAG)**: Large Language Models are prone to *hallucinations*, i.e. instances in which the model produces factually incorrect or nonsensical information, especially when handling long contexts. This behavior is particularly dangerous when applied to uses in certain professional fields such as law or healthcare. RAG addresses this by conditioning the model's output on auxiliary information, in the form of a set of retrieved sources drawn from an external corpus, augmenting the original prompt with information not necessarily memorized in the LLM. This is aimed at helping the model generate more accurate and transparent answers at inference time.

  A RAG system consists of two fundamental components: a *retriever* (responsible for invoking an Information Retrieval system and feeding its results to the generator) and a *generator*(Cuconasu et al., 2024).

  Given a query q, the **retriever** selects a subset of relevant passages or documents $\{d_1, d_2, \ldots, d_n\}$ from the corpus to allow the reasoner to answer q correctly. It utilizes an embedding process to generate corresponding embeddings $\vec{q}$ for the query and $\vec{d_i}$ for each passage in the corpus(Cuconasu et al., 2024):

  $$\vec{q} = Encoder_q(q); \vec{d_i} = Encoder_d(d_i)$$

  Similarity $s(q, d_i)$ is then computed to quantify the relevance of each passage to the query.
  A **generator**, or reasoner, component is then in charge of synthesizing an answer using the top-rated documents. Typically this is implemented via an LLM, and this project is no exception.

  In particular, in the context of this project RAG ensures that every discharge recommendation is traceable to a specific guideline excerpt, enabling the reviewing physician to verify each suggestion before delivering it to the patient.

- **Sentence Tokenization**: Sentence tokenizers, such as the Punkt algorithm, are unsupervised model that automatically recognize abbreviations, acronyms, and sentence boundaries that split text into sentences. They are not language specific and can handle punctuation marks more intelligently than applying simple rule-based systems such as "a sentence ends when there is a period"(PUNKT).
  The algorithm is trained to examine a specific text corpus for clues about what constitutes a sentence boundaries(punkt module).

  They are used in this project to ensure that chunk boundaries always fall between complete sentences, avoiding the mid-sentence or mid-word splits that a fixed-length character-based approach would produce.

- **Embedding Models**: Word embedding is a real-valued vector representation of words by embedding both semantic and syntactic meanings obtained from unlabeled large corpus(Wang et al., 2019). Embedders play a central role in machine learning, and their use is prominent in fields such as Natural Language Processing or Computer Vision.
  Embedding models map textual inputs into dense vector representations in a high-dimensional space $\mathbb{R}^d$, where semantic and conceptual similarity between texts corresponds to geometric proximity between their vectors(Wang et al., 2019). Given an embedder E and a text t:

  $$\mathbf{v} = E(t) \in \mathbb{R}^d$$

  In a retrieval setting, it's common practice for the corpus documents and the user's query to be embedded using the same model, to ensure consistency of results. Relevant passages are identified by computing a similarity score between the embeddings of the query-passage couple, with the goal of finding the minimum distances in the vector space (faiss.ai):

  $$j = argmin_i ||q - d_i||$$

  Embedding models must be specifically trained with a contrastive objective that pulls semantically related texts together while also pushing unrelated texts apart in the resulting embedding space.
  Domain-specific embedding models, trained or fine-tuned on specific literature, may be able to capture the nuances of domain-specific terminology more effectively than general-purpose models, making them better suited for distinguishing between superficially similar but distinct concepts. However, it's still common for broad embedders with an elevated number of parameters to outperform smaller domain-specific models.

- **Named Entity Recognition (NER)**: Named Entity Recognition is the process of locating a word or a phrase that references a particular entity within a text(Sharnagat, 2014). NER involves two tasks, which is firstly the identification of proper names in text, and secondly the classification of these names into a set of predefined categories of interest.
  While there are different classes of NER models (Mansouri et al., 2008), they all follow the same logical process: given a sequence of words $(x_1, x_2, \ldots, x_n)$, the model assigns a label $y_i \in \mathcal{Y}$ to each of them, where $\mathcal{Y}$ is the set of entity categories (e.g. DISEASE, MEDICATION, PROCEDURE) augmented with a non-entity label.

  In the clinical domain, NER models may be trained to recognize a plethora of entities such as diseases, medications, symptoms, and diagnostic procedures within unstructured clinical narratives.
  In this system, NER serves as the first step in query decomposition: rather than embedding the entire clinical note as a single query, the system extracts distinct entity types and uses them to construct targeted sub-queries, each focusing on a specific clinical dimension.

- **TF-IDF**: Term Frequency-Inverse Document Frequency is a sparse retrieval method commonly used to weigh each word in the text document according to how unique it is relative to the broader corpus. TF-IDF works by computing a score w between a query term t and a document d, based on the weight of t in d:

  $$\text{tf-idf}_{t,d} = \text{tf}_{t,d} \times \text{idf}_t$$

  Where the term frequency is the count of t in d, and the inverse document frequency is inversely proportional to the number of documents t appears in:

  $$\text{idf}_t = \log \frac{N}{\text{df}_t}$$

  Terms that appear in many documents receive a low IDF score, while terms that are concentrated in a small number of documents receive a high score.
  For the purpose of this project, Term Frequency should be computed on the input clinical note and multiplied to the Inverse Document Frequency computed on the document corpus. However, since it is assumed that the number of times a term appears in the clinical note is not related to its importance in the diagnosis, $\text{tf}_{t,d}$ is a binary entity of value 1 if the term is in the clinical note, 0 otherwise.

- **Top-k Retrieval**: Given an embedded query $\mathbf{q}$ and an index containing M embedded document vectors, top-k retrieval selects the k documents whose vectors are most similar to the query:

$$\text{Top-}k(\mathbf{q}) = \underset{S \subset \{1,\ldots,M\},\ |S|=k}{\arg\max} \sum_{i \in S} \text{sim}(\mathbf{q}, \mathbf{d}_i)$$

While there exist many ways to compute vector similarity, a very common method, also implemented in this project, consists in the usage of **cosine similarity** computed between the query vector $\mathbf{q}$ and each document vector $\mathbf{d}_i$(Steck et al., 2024):

$$\text{sim}(\mathbf{q}, \mathbf{d}_i) = \frac{\mathbf{q} \cdot \mathbf{d}_i}{\|\mathbf{q}\|\,\|\mathbf{d}_i\|}$$

Since vectors in FAISS are normalized, this is equivalent to computing **dot-product** similarity $\text{sim}(\mathbf{q}, \mathbf{d}_i) = \mathbf{q} \cdot \mathbf{d}_i$

Top-k retrieval is designed to be fast and recall-oriented, casting a wide net over the corpus to ensure that all potentially relevant chunks are included in the candidate pool. The trade-off is precision; some of the retrieved candidates may be only superficially related to the query. In fact, the quality of this retrieval method is highly dependent on the quality of the precedent embedding process.

- **Cross-Encoders and Reranking**: The two-stage retrieve-then-rerank paradigm exploits a fundamental trade-off between efficiency and accuracy in neural information retrieval. In the first stage, an embedding model (bi-encoder) embeds queries and documents independently, enabling fast retrieval via vector similarity, but sacrificing the ability to model fine-grained interactions between query and document tokens. Results produced in the retrieval stage by the top-k algorithm can be noisy or irrelevant, and are highly dependent on the quality of the embedding.

In the second stage, a cross-encoder takes the query and each candidate document as a single concatenated input and produces a scalar relevance score(Reimers and Gurevych, 2019):

$$s(q, d_i) = \text{CrossEncoder}([q; d_i]) \in \mathbb{R}$$

The concatenated input allows for more granular similarity analysis between the query and the document, yielding substantially more accurate relevance estimates than independent encoding(Reimers and Gurevych, 2019). However, the computational cost scales linearly with the number of candidates, since each pair requires a full forward pass, making it prohibitive for the full corpus(Geigle et al., 2022).

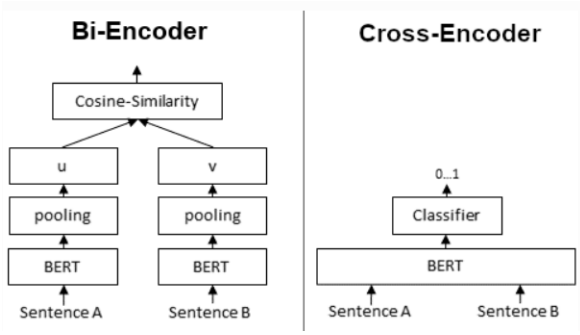

Figure 1: Architecture comparison

3 System Architecture

The system is organised as two separate but connected pipelines, as illustrated in Figure 2.

- **Guideline ingestion pipeline**, responsible for processing, embedding and handling the vector storage of the medical guideline corpus.
- **Query processing and generation pipeline**, which takes as input the clinical note and outputs the discharge recommendations at inference time.

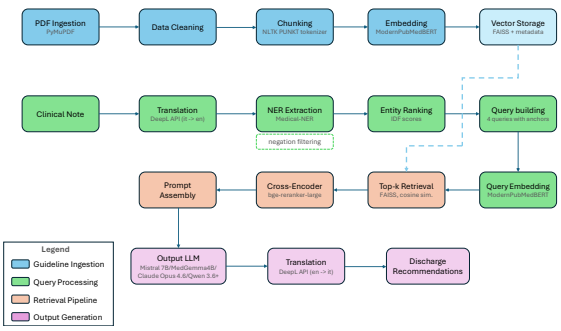

Figure 2: System architecture

3.1 Guideline Ingestion Pipeline

The guideline ingestion pipeline runs offline and only needs to be executed when the corpus is

updated. Technically, it could even be executed only once, but guidelines are updated over the years, so a periodical update of the corpus would be good practice.

European medical guidelines, published as PDF documents and downloaded manually into the local machine, are first parsed using **PyMuPDF** to extract raw text with page-level metadata. The extracted text then undergoes a cleaning stage that aims to remove pages that are purely noise for the purpose of the model: index pages, abbreviation summary pages, bibliography and references. This is a particularly important step of the process, since those pages are condensed with clinically relevant terms that don't convey useful information but can produce high cosine similarity scores and pollute retrieval.
Pages are then processed and split into raw lines, where a line-level cleaning process is performed to remove reference numbers and replacement characters.
The cleaned lines are merged into sentences using a NLTK's **PUNKT tokenizer** trained on the whole guideline corpus. Then, a third and final cleaning step (sentence-level) eliminates sentences that are clearly Figure/Table captions or illegible summary tables.

Sentences are merged into overlapping chunks, and document-level metadata (source guideline, page range) is preserved for each chunk. Using PUNKT to form the sentences ensures that chunk boundaries always fall between sentences rather than within them, avoiding the mid-sentence or even mid-word splits that a fixed-length character-based approach would produce.
Finally, each chunk is embedded using **ModernPubMedBERT**, a specialized medical embedding model fine-tuned on biomedical (PubMed) data(ModernPubMedBERT), and stored in a **FAISS** flat inner-product index, alongside its metadata, in a pickle file.
IDF scores for the whole corpus are also computed on the resulting chunks and stored within a joblib file.

Figure 3: Guideline ingestion pipeline architecture

3.2 Query Processing and Generation Pipeline

At inference time, the system receives a clinical note written in Italian as a raw *.txt* file. The following handling of the data was made with the assumption that the model would be integrated into the normal program physicians use to produce the clinical documents. Therefore, it is assumed that the text file would always include the standard sections produced during an ER visit: medical history, objective examination, clinical diary, diagnosis, and prognosis.
The note is translated from Italian to English via the **DeepL API**, using a custom glossary to ensure consistent translation of clinical section headers and a selection of common medical terms and abbreviations. The section headers are then used to transform the raw string extracted from the text file into a dictionary that can then be processed in the rest of the pipeline.

Early experiments showed that embedding the entire clinical note as a single query led to ineffective retrieval, as the large amount of information caused the retriever to return generic, cross-domain chunks rather than targeted content. To address this, a decision was made to decompose the clinical note into multiple sub-queries, each focusing on a distinct aspect of the clinical treatment, and perform retrieval independently for each.
The decomposition begins by processing the translated note with a **Named Entity Recognition** model (*Clinical-AI-Apollo/Medical-NER*), which extracts medical entities across three categories: diseases, medications, and diagnostic procedures. A clause-aware negation filter then removes entities that appear in negated contexts (e.g. *denies chest pain*, *negative pregnancy test*), preventing the retrieval of guideline chunks that would not be relevant to the patient's actual clinical picture.
The surviving entities are scored using a cross-corpus TF-IDF scheme, where inverse document frequencies are derived from the guideline corpus itself. This allows the system to prioritise entities that are statistically distinctive within the guideline vocabulary. For instance, terms like X-ray that are clinically relevant but not indicative of a specific condition, would get penalized in order to focus on more specific terms that can lead to the retrieval of relevant chunks.
Each sub-query is then assembled by combining

only the top-ranked entities with the patient's diagnosis and a set of topic-specific anchor terms (e.g. for diseases: *diagnosis comorbidities risk factors management*) to guide retrieval. A fourth, broad, sub-query containing the diagnosis, anchor terms, age bracket and gender of the patient is also created.

Each query is then embedded with the same **ModernPubMedBERT** encoder used during ingestion, and matched against the FAISS index via cosine similarity to retrieve the top-k candidate chunks. Candidates from all queries are merged and duplicates, if present, are removed.
The merged candidate pool is then reranked by a **Cross-Encoder** (*BAAI/bge-reranker-large*), which scores each query-chunk pair independently. This does create some latency, being the most time-consuming part of the retrieval and taking $\sim$2 seconds. However, it drastically improves the output quality, so it is considered an acceptable trade-off, especially in light of the fact that it is still much faster than the LLM output generation phase.
The final ranking is determined by max-aggregation across queries, supplemented by a category leader boost, a primary guideline boost, and text-level deduplication to remove near-identical chunks.

The top ranked chunks are then assembled into a structured prompt together with the clinical note and a set of task instructions. Based on the extracted prognosis field, the system detects whether the patient is being hospitalized, and therefore handed over to the care of a different medical team (a discharge document is still produced in these cases, but recommendations are different as they do not contain therapies), or discharged home, and selects the appropriate prompt instructions and template accordingly.

Finally, the prompt is passed to a **Large Language Model** for generation and the English output is translated back to Italian through **DeepL**.
The output, exported in a *.txt* file, is organized into two sections: the first consists of a numbered lists of recommendations, ready to be inserted into the discharge document, and the second contains a brief justification for each recommendation.

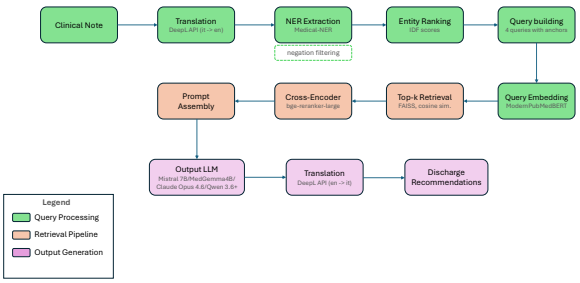

Figure 4: Guideline ingestion pipeline architecture

4 Implementation

This section outlines the concrete implementation of the pipeline, including the specific models and tools selected, the decisions made during development, and the rationale behind them.

4.1 Data Selection, Ingestion and Processing

The guideline corpus consists of European medical guidelines published as PDF documents by independent yet official and recognized organizations such as the European Society of Cardiology (ESC), the European Hematology Association (EHA), and other European medical societies.
This introduces the first hardship encountered in the construction of the system: data is not directly downloadable from a single, recognized source, but scattered in various forms across many different sites managed by independent medical organizations. They are not readily available for download and use, and do not follow a common structure or philosophy, making rule-based cleaning of the files less straightforward and a more time-demanding operation.
After obtaining the guidelines, PDF parsing is performed with PyMuPDF, a Python library which extracts raw text while preserving page-level positional information used to associate each chunk with its source document and page range. In the ingestion phase, positional-level information of text blocks is used to filter out the page margins, allowing to immediately discard footnotes, page numbers and other noisy text while only keeping the relevant parts of each page.

Still, raw PDF extraction ingests noise alongside useful data, and a further process to remove the latter before text can be meaningfully embedded is necessary. The cleaning pipeline applies the following operations:

- **Page-level cleaning**: Alongside the aforemen-

tioned page-border removal, three operations are executed:

  - *Index Pages Removal*: a multi-heuristic weighted scoring system identifies and excludes pages that satisfy certain conditions. The heuristics are keyword detection (0.3 weight), lines ending with page numbers (up to 0.4, sufficient condition if more than half raw lines end with a number) and dot or dash leaders (up to 0.2), with a classification threshold of 0.4.
  - *References Pages Removal*: a big component of guidelines, in some cases References section constitute more than 30% of the total pages. They are identified by a high number of reference-common patterns such as "doi", "http", "2021;28:".
  - *Abbreviation Pages Removal*: Eliminates the page if there is a high number of raw lines that start with an abbreviation pattern (a mostly uppercase short word that constitutes a sentence on its own).

- **Line-level cleaning**: Before merging the sentences, another layer of cleaning is applied on the raw lines:
  - *Reference numbers removal*: inline citation markers are stripped to prevent them from polluting the semantic content of chunks, while still setting rules to prevent the elimination of meaningful numbers.
  - *Garbled line filtering*: lines containing a high proportion of non-alphabetic characters or malformed Unicode sequences are discarded.

- **Sentence-level cleaning**: after lines are merged into sentences with the aid of an instance of NLTK's Punkt sentence tokenizer trained on the whole guideline corpus, a final level of cleaning is applied:
  - *Summary removal*: aims to remove sentences that come from summaries or tables that were extracted without significant context, they are simply a collection of semantically meaningful terms (polluting retrieval) without any significance to them. These extracted sentences contain a lack of punctuation over elevated number of characters or an excess of semicolons, and are flagged if those conditions are met.
  - *Caption removal*: Removes sentences that start with keywords such as "Figure" or "Recommendation Table".

It is important to underline that this is an approach that aims to remove only sections of text that are obviously noise, while leaving room for dubious page to pass through the next steps. The cleaning philosophy is that the cost of a false positive (eliminating useful information) is much higher than the cost of a false negative.

After cleaning and sentence merging, chunks are constructed by concatenating consecutive sentences up to a target character length (1200, set heuristically), with a 2 sentence overlap between adjacent chunks to preserve context across boundaries. Each chunk retains metadata including the source document and the page range it spans.

4.2 Embedding and Vector Storage

Each cleaned and chunked guideline passage is embedded using ModernPubMedBERT, a biomedical sentence encoder, chosen as the final model after evaluating several candidates:

- *ModernPubMedBERT*(ModernPubMedBERT): A specialized medical embedding model, fine-tuned from Clinical ModernBERT using InfoNCE contrastive learning on PubMed title-abstract pairs. Weighing only 600 Mb (an important parameter, since the model needs to be loaded in the VRAM memory not only for chunk embedding, but query embedding as well), it performed very well and was ultimately chosen as the final model.

- *Bio_ClinicalBERT*(BioClinicalBERT): Another ClinicalBERT based model trained on notes from MIMIC III, its performances were sub-par, producing effectively random retrieval results .

- *MedEmbed-large*(MedEmbed): An embedding model fine-tuned specifically for medical and clinical data, obtained results that were extremely similar to ModernPubMedBERT, but due to its greater size (1.3 Gb) was ultimately discarded.

- *MTEB Leaderboard Models*(MTEB): Consideration was given to the models that were

leading the Massive Text Embedding Benchmark Leaderboard in the Healthcare Domain category, such as *voyage-4* (leading the leaderboard but not publicly available), and *Octen-Embedding-8B*. However, these powerful models are very heavy (weighing over 8Gb) and thus deemed unsuitable for the project.

The resulting embedded vectors are L_2-normalized and stored in a FAISS (faiss.ai) `IndexFlatIP` (flat inner-product index), which computes exact nearest-neighbour search via dot product (as stated earlier, equivalent to cosine similarity when vectors are normalized). Chunk metadata (source document, page range, full text) is serialized alongside the index in a pickle file.
A TF-IDF vectorizer, fitted on the full chunk corpus and containing the IDF weights, is also persisted via `joblib` for use in the query construction stage.

4.3 Translation Layer

The entire retrieval and generation pipeline operates in English, as the guideline corpus is composed exclusively of English-language documents. However, since the clinical input originates in Italian, a translation step is required at both ends of the pipeline.

The Italian-to-English translation is performed via the DeepL API (DeepL), configured with a custom glossary that maps Italian clinical section headers (*Anamnesi*, *Esame obiettivo*, *Diario clinico*, *Diagnosi*, *Prognosi*) to fixed English equivalents. This ensures that the downstream field extraction logic, which relies on regex-based section detection, operates on consistent header text regardless of how DeepL might otherwise choose to translate these terms.
While the DeepL model performs extremely well in the medical field, it is not a clinical-specific model. Therefore, the glossary is extended with common medical specific terms and abbreviations that DeepL is not able to pick up on its own (e.g. *TC* is mapped to *Body Temperature*, *Giordano* is mapped to *CVA Tenderness*).

Two alternative strategies to this translation approach were considered but subsequently discarded.

- Utilizing a local model for translation, the original approach, either led to low-quality results (*Helsinki-NLP/opus-mt-it-en*(Helsinki Opus)) or to unbearable computational effort (*google/madlad400-3b-mt*(Google Madlad 400-3b), weighing almost 12Gb and taking more than 500 seconds to translate a clinical note, *facebook/seamless-m4t-v2-large*(SeamlessM4T v2) was discarded for the same reason).

- Adopting a multilingual embedding model was rejected because multilingual encoders tend to produce less precise representations than monolingual models for domain-specific retrieval, particularly with specialized medical terminology.

For the reverse translation (English output to Italian), two approaches were considered. The default uses DeepL for consistency and reliability. An alternative two-prompt approach, sending the English output back to the same LLM with a translation instruction, which produces more natural-sounding Italian at the cost of an additional API call, is still deemed worthy of consideration at the time of writing, though the code has not been implemented. While LLM self-translation is viable for larger API-served models, DeepL is certainly to be preferred for small local models.

4.4 Query Construction

Early experiments demonstrated that embedding the full clinical note as a single query produced ineffective retrieval: the large amount of clinical information diluted the semantic signal, causing the retriever to return generic, cross-domain chunks rather than targeted guideline content. To address this, a decision was made to separate the clinical note into multiple focused sub-queries, each targeting a distinct clinical dimension.
A first attempt at separating the document's natural entities (clinical diary, medical history, etc.) and combining them with specific anchor and the broad diagnosis proved equally unsuccessful, as the amount of information was still very large and led to generic retrieval. Therefore, a more complex approach was adopted.

4.4.1 Named Entity Recognition and Negation Filtering

The translated clinical note is processed by *Clinical-AI-Apollo/Medical-NER*(Medical-NER),

a transformer-based NER model fine-tuned on clinical text, which extracts entities across sixteen categories, among which four were initially deemed semantically relevant and used for sub-query construction: DISEASE_DISORDER, MEDICATION, SIGN_SYMPTOM, and DIAGNOSTIC_PROCEDURE. A second model, *OpenMed/OpenMed-ZeroShot-NER-Disease-Large-459M*(OpenMed), was also considered but discarded due to its inability in effectively identifying categories different from DISEASES.

Still, even with a solid NER model, not all extracted entities are suitable for query construction. A clause-aware **negation filter** was implemented to examine the textual context surrounding each entity to determine whether it appears in a negated form; these terms, while clinically relevant, would only serve to poison retrieval by introducing conditions that were already excluded by the physician. The filter operates by inspecting a window of characters preceding the entity (up to the nearest clause boundary, identified by periods, commas, colons, or semicolons) for negation patterns such as *no*, *not*, *denies*, *absent*, *without*, and *ruled out*. A forward-looking check also detects post-entity negation (e.g. *Blumberg test negative*). Entities identified as negated are excluded from query construction, preventing the retrieval of guideline content for conditions the patient does not present.

4.4.2 Entity Type Strategy

Through iterative experimentation, a strategy for the selection and usage of entity types was developed:

- **Active types** (DISEASE_DISORDER, MEDICATION): these entities generate their own retrieval queries, as they produce focused, guideline-specific retrieval results.
- **Semi-Active type** (DIAGNOSTIC_PROCEDURE): the top two entities by TF-IDF score are inserted into their clinical context (the whole sentence they appear in is retrieved rather than only the entities themselves) and used to form a query that is used exclusively in the reranking stage, where they help surface drug dosing tables and treatment protocol chunks without polluting the FAISS retrieval pool.
  While selecting only two entities may seem restrictive, empirical evidence showed that most of the entity type's elements are common procedures (e.g. ECG, X-rays) that are performed on most of the patients, and are therefore not relevant in the retrieval of domain-specific guidelines.
- **Dropped type** (SIGN_SYMPTOM): excluded entirely from query construction. Symptoms such as *alert*, *oriented*, and *collaborating* are ubiquitous across clinical notes and produced no discriminative retrieval signal in any tested configuration.

4.4.3 TF-IDF Ranking

Within each active entity type, extracted entities are ranked by their IDF score computed over the guideline chunk corpus. For multi-word entities (e.g. *atrial fibrillation*), the entity's score is defined as the maximum IDF across its constituent terms. Entities whose tokens do not appear in the guideline vocabulary receive a score of zero and are discarded. A maximum term count, determined heuristically, per category prevents low-relevance entities from diluting the query.

4.4.4 Query Assembly

Each sub-query is assembled by concatenating three components: the top-ranked entities for the category (or in the case of DIAGNOSTIC_PROCEDURE, their context), the patient's diagnosis (extracted from the clinical note), and a set of domain-specific anchor terms. The anchor terms are fixed strings designed to steer the embedding toward recommendation-bearing regions of the guideline vector space and determined through heuristic experiments. For example, the medication query uses the anchor *therapy recommendations pharmacological management clinical scores diagnostic evaluation*.
A broad diagnosis-level query is always included, combining the diagnosis with generic discharge-oriented terms and, where applicable, demographic qualifiers such as age bracket (*elderly*, *paediatric*, *neonatal*) and gender.

4.5 Information Retrieval and Reranking

4.5.1 Top-*k* Retrieval

All sub-queries, with the exception of the DIAGNOSTIC_PROCEDURE one, are embedded using the same ModernPubMedBERT encoder used during indexing. The resulting query vectors are then matched against the FAISS index, and

the top-k candidates ($k = 10$) are retrieved for each query. Candidate sets from each queries are merged and deduplicated by FAISS index identifier, producing a unified pool that typically contains 25-30 unique chunks.

MMR was initially implemented with the idea of reducing the number of candidates passed to the Cross-Encoder while promoting diversity among retrieved results. However, it was later removed after improvements to the multi-query strategy achieved this goal without introducing an additional point of failure.

4.5.2 Cross-Encoder Reranking

The merged candidate pool is reranked using *BAAI/bge-reranker-large*(bge-reranker), a cross-encoder model that was selected over *ncbi/MedCPT*(MedCPT) after comparative testing showed consistently stronger relevance discrimination across clinical cases. This result was somewhat surprising since MedCPT is a medical specific model, but the quality of the bge-reranker was still superior due to its higher number of parameters.

All query-chunk pairs are scored in a single batch call to the cross-encoder's `predict()` method, producing a score matrix of dimensions $Q \times N$ (number of queries $\times$ number of candidates). The final relevance score for each chunk is determined by max-aggregation across queries, ensuring that a chunk which is highly relevant to any single clinical dimension is retained even if it scores poorly against other sub-queries. Max-aggregation was preferred over mean-aggregation because the latter favors broadly generic chunks that score moderately across all clinical dimensions without being strongly relevant to any, precisely the cross-domain contamination the multi-query strategy was designed to avoid.

Scoring every query-chunk pair through the cross-encoder is a computationally expensive process, especially relative to bi-encoder retrieval. However, the added latency ($\sim$2 seconds) remains well below that of any generation method, and the improvement in retrieval precision justifies the cost.

Three additional scoring adjustments are applied over the Cross-Encoder scores after max-aggregation:

- **Category leader boost**: the top-scoring chunk for each individual query receives a 10% multiplicative bonus (non-cumulative), rewarding chunks that are the best match for a specific clinical dimension, without imposing its presence if it is not a particularly relevant chunk.
- **Primary source boost**: if a specific guideline is retrieved in at least 35% of the chunks in top-k, it is considered the primary information source and all its chunks are given a 15% score boost.
- **Text deduplication**: when two chunks share more than 80% token overlap, the lower-scoring chunk is removed to avoid presenting near-identical content to the language model.

The top N chunks after reranking ($N = 10$) are selected for inclusion in the generation prompt.

4.6 Output Generation

4.6.1 Prompt Design

The generation prompt is assembled from four blocks: a role definition, the full clinical note, the retrieved guideline excerpts (each labelled with a citation identifier (`[E#]`), source document, and page range), and a task instruction block. The system detects whether the patient is being discharged home or admitted to hospital by inspecting the extracted prognosis field for hospitalization-related keywords, and selects one of two prompt templates accordingly.

For patients discharged home, the prompt instructs the model to produce two sections: a numbered list of concise, directly actionable recommendations addressed to the patient (without inline rationale or pathophysiological explanation), followed by a clinical references section where each recommendation is mapped to its supporting excerpt citation and a brief justification for the reviewing physician. Specific formatting rules enforce medication dosing completeness (drug name, dose, route, frequency), prohibit speculation about future treatment decisions, require strong advisory tone for lifestyle recommendations, and mandate standardized phrasing for specialist referrals and emergency return criteria.

For patients admitted to hospital, a separate prompt template produces a smaller handover

note (admission statement, informed consent documentation, and team handover) followed by the same clinical references structure. This template explicitly prohibits prescribing medications or follow-up, as these responsibilities belong to the admitting team.

In both cases, the model is specifically instructed to completely ignore any previous knowledge and base all recommendations exclusively on the provided guideline excerpts, limiting the risk of hallucinations as much as possible.
Examples of resulting prompts are included in the Appendix.

4.6.2 Language Models

The pipeline supports multiple generation back ends to enable comparative evaluation. This is the most flexible stage of the system and a final definitive decision over a definitive model or generation backend has been willingly omitted to allow for different needs, budgets and privacy constraints. Four models were tested in the experimental runs:

- *Mistral 7B Instruct* (Mistral7B) (quantized, local): a 6-bit quantization of the instruct fine-tuned version of the Mistral-7B-v0.3. It represents a lightweight baseline requiring approximately 6GB of VRAM, used to establish a lower bound on output quality achievable with relatively minimal hardware.

- *MedGemma 4B-it* (MedGemma) (MedGemmaQuantization) (quantized, local): an 8-bit quantization of the instruction-tuned Gemma 3 variant, fine-tuned for medical text and image comprehension.
  Its inclusion serves as an experiment in whether domain-specific pretraining improves output quality in a task that, despite being encased in a medical domain, is primarily instruction-following and text production, rather than proper clinical reasoning. Requiring approximately 5GB of VRAM, it remains deployable on modest hardware.

- *Qwen 3.6 Plus (free)* (API via OpenRouter (Openrouter)): a high-range model provided for free via Openrouter. However, the free tier leads to much longer inference times, even though output quality is elevated.

- *Claude Opus 4.6* (API via OpenRouter (Openrouter)): the most capable model tested, used as an upper bound for generation quality. Generation costs about 0.03$ per document.

The generation backend is configurable, allowing the same retrieval pipeline and prompt to be evaluated across different models without any changes to the upstream components. Local models can be loaded either via the `Transformers` library or the `Llama-cpp` one (for GGUF quantized versions), while the API is designed to work with the Openrouter API.

5 Evaluation

Evaluating the quality of automatically generated clinical documents presents challenges that are inherent to the nature of the task. In fact, no standardized metric, analogous to ROUGE for summarization or BLEU for machine translation, exists for discharge recommendation generation. Even if reference-based metrics were applicable, they would be poorly suited to this domain: clinical texts are dense with synonyms and equivalent formulations (e.g. *aspirin* and *analgesic* are clinically analogous but would be penalized as mismatches), and one of the explicit goals of this system is to surface recommendations that were omitted in the original, manually written discharge notes, meaning that a high-quality output may deliberately diverge from its reference.

A rigorous evaluation would require multiple domain experts to independently score a large volume of generated documents against clinical standards, but this was not feasible within the budget of this project. The evaluation presented here is therefore primarily qualitative, conducted across nine clinical cases that were authored by a physician with specific instructions to produce recommendations that were as legible as possible, and span a range of diagnoses and clinical complexities.
Two dimensions are assessed independently: the quality of the retrieved guideline excerpts, evaluated using an LLM-as-a-judge approach, and the quality of the generated discharge recommendations, assessed through manual comparison with the original physician-written recommendations.

5.1 Retrieval Evaluation

To assess retrieval quality, an LLM-as-a-judge approach was adopted. For each of the nine clinical cases, the final set of ten retrieved guideline excerpts was evaluated by three different large lan-

guage models (Claude Sonnet, ChatGPT Base and Gemini 3) on the basis of clinical relevance to the patient's diagnosis and actionability for discharge recommendation generation.
The following prompt was given to each of the LLMs:

> *In each of the attached text files, you are given a clinical note produced from a hospital and ten guidelines excerpts that were extracted using an IR process designed to retrieve clinically relevant chunks. These chunks will be later used as the sole source of information in a RAG system that generates clinical recommendations to the patient based on the contents of the clinical note. You are tasked with scoring each case on a ten-point scale, where a score of 10 indicates that all retrieved excerpts are directly relevant and actionable, and a score of 0 indicates complete retrieval failure. Provide a clear explanation of the score for each case.*

Retrieval files were passed in batches of 3 to avoid excessively long contexts which would risk reducing the judgment quality. Scores from the three models are reported in Table 1.

With the exception of the Pneumonia case, whose poor performance is attributable to a gap in the guideline corpus (see Limitations section), retrieval quality is generally satisfactory, with strong results observed in several diagnoses.
Importantly, a score of 4 should not be interpreted as an outright failure: it indicates that 4 out of the 10 retrieved chunks are clinically relevant and usable for downstream generation. Although this means some noise will also be retrieved, a sufficiently capable LLM should still be able to produce a meaningful output, even when only a subset of the retrieved evidence (e.g., 4-5 chunks) is informative, as analyzed in the Output Evaluation section. Moreover, some of the low-quality retrieved chunks come from the right guidelines, but are retrievals of dense summary tables, meaning even a less powerful LLM is normally able to discern them as noise without risking pollution from low-quality information.

5.2 Output Evaluation

The generation quality was evaluated by comparing the outputs of four LLMs on the same set of clinical cases, using the same retrieval results and prompt template. This isolates the effect of model capability from the upstream pipeline components.

The four models tested were: Mistral 7B Instruct (quantized, local), MedGemma 4B-IT (quantized, local), Qwen 3.6 Plus:free (API), and Claude Opus 4.6 (API).
It is worth noting that, due to budget constraints, no powerful model was available for evaluation in their local version. While this should not influence the commentary on the accuracy of the output production, it would have most certainly influenced the production time.
The more powerful models (Qwen 3.6 Plus and Claude Opus 4.6) were prompted via the Openrouter.ai API, with the rest of the pipeline running on an NVIDIA RTX 4050 laptop with 6GB VRAM available. The two local models were run through the free tier of Google Colab, connecting to the T4 GPU hardware accelerator with the 07.2025 runtime version (CUDA 12.4).

Evaluations were performed manually with the aid of an Emergency Room physician, using the original clinical recommendations as standard for comparison.
Two specific cases are discussed more in detail in the Appendix A section.
All outputs from the various models are available in the GitHub repository, inside the `output [MODEL_NAME]` directories.

Each output was assessed along four qualitative axes:

1. **Clinical accuracy**: whether recommendations were grounded in the retrieved excerpts without hallucinated drug names, dosages, or clinical facts.

2. **Instruction adherence**: whether the model followed the two-section output format, avoided inline rationale in Section 1, and respected all formatting constraints (no headers, no sub-bullets, no citations in Section 1).

3. **Conciseness**: whether recommendations were direct and actionable, without unnecessary

Clinical Case	Retrieval Scores (0–10)		
	Claude (Sonnet)	**ChatGPT**	**Gemini (Thinking)**
Atrial Fibrillation	8	8	10
Kawasaki Disease	8	9	9
Gastroesophageal Reflux Disease	4	5	8
Deep Vein Thrombosis	6	6	7
Cellulitis / Infective Endocarditis	9	8	10
Pneumonia[a]	1	2	0
Acute Appendicitis	6	9	8
Malleolar Fracture	3	7	2
Extrauterine Pregnancy	3	7	3
Average	6.22 (6.625 excluding Pneumonia)		

[a] Lack of relevant pneumonia guidelines in the retrieval corpus.

Table 1: Average retrieval quality scores across the nine evaluation cases, assessed via LLM-as-a-judge. Original chats can be accessed through the following links: Claude, ChatGPT, Gemini.

Low scores are due to a lack of relevant pneumonia guidelines in the retrieval corpus.

verbosity, speculative commentary, or redundant phrasing.

4. **Speed**: whether the latency was acceptable, and if it can be considered an improvement over manual drafting of clinical recommendations.

5.2.1 Mistral 7B Instruct

The model was selected as a lightweight alternative that can be executed even with minimal hardware requirements. It is imported locally in its 6-bit GGUF quantization version.

- **Clinical Accuracy**: Overall, the quality of the generated output is insufficient.
  While the model is occasionally able to produce clinically relevant recommendations, its performance is undermined by frequent hallucinations.
  Some minor hallucinations, such as immuted lifestyle advice, may be tolerable, even though they deviate from the instruction to only include lifestyle recommendations that modify the patient's daily behavior. For instance, a recommendation to stop smoking would be welcomed, while one to maintain a healthy lifestyle would be redundant in an official clinical document.
  More concerning, however, are clinically unsafe suggestions (e.g. explicit encouragement to continue oral hormonal therapy in the context of thrombosis), which would be outright dangerous if not reviewed and corrected by the hypothetical supervising physician utilizing the system.
  It also writes that the patient has been prepped for operation every time the prognosis is hospitalization, even when the patient doesn't require surgical intervention.

- **Instruction Adherence**: While the model followed the two section format consistently, and avoided explaining the rationale in-line, the phrasing used to provide recommendations to the patient sounded unnatural at times. For instance, it has the tendency to add "Strongly recommended" as a standalone sentence at the end of the recommendation, sounding aggressive.
  Sentence format would need minor adjustments before being presented to the patient.

- **Conciseness**: The output is concise and presentable to the patient as is.

- **Speed**: The model generates recommendations with an average latency of approximately 37 seconds per case. When combined with the $\sim$2 seconds required for the information retrieval step, the total processing time amounts to roughly 39 seconds per output.

  While not optimal, this still represents a potential improvement over manual drafting of discharge recommendations in clinical practice.
  Note: Models loading time ($\sim$8 seconds) is

excluded from these measurements. This assumes a deployment setting in which models are initialized once at system startup and remain loaded during operation.

Summary Mistral 7B Instruct demonstrates limited reliability for clinical use. Although it can occasionally produce relevant recommendations, its outputs are frequently affected by hallucinations, including clinically unsafe suggestions that necessitate careful supervision. Its latency performances are acceptable, but are not enough to offset the clinical accuracy.
Overall, the model is not production ready.

5.2.2 MedGemma 4B-IT

This is the smallest model evaluated (4B parameters), selected to assess whether domain-specific training in clinical reasoning provides measurable benefits. However, within the current system design, where the diagnosis is given, the task is primarily one of instruction following and structured text generation rather than diagnostic reasoning. The experiment was therefore intended to evaluate whether domain specialization could still offer an advantage under these conditions.

- **Clinical Accuracy**: The generated output is, quite simply, inadequate.
  The model exhibits a pronounced tendency toward hallucination, and the alignment between recommendations and their supporting rationale is wildly inconsistent. In several cases, the explanatory section does not match the number or content of the recommendations. Rather than brief justifications, the model often reproduces large portions of retrieved text that at times are entirely unrelated to the original recommendation it should explain.

- **Instruction Adherence**: The model struggles to reliably follow the required two-section structure and, in multiple instances, produces duplicated outputs.

- **Conciseness**: While outputs are generally concise, the model occasionally introduces redundant or unnecessary recommendations. In extreme cases, it can commit repetition errors, including one instance where the same recommendation was repeated over 100 times.

- **Speed**: The model generates outputs with an average latency of approximately 78 seconds per case. Including the $\sim$2 seconds required for information retrieval, total processing time reaches roughly 80 seconds per output. This is at the limit for convenience when compared to manual drafting, especially when accounting for the additional effort needed for review and correction.

Summary Domain-specific pretraining in clinical reasoning does not compensate for the model's limited capacity in this setting. On the contrary, the combination of small model size and relatively long input context (approximately 5000 characters) proves to be too much to handle. The model presents no upsides, therefore it does not present a viable option for deployment.

5.2.3 Qwen 3.6 Plus (free)

This model was accessed via the OpenRouter API and selected as a high-capacity, cost-free alternative, with claimed performance comparable to state-of-the-art systems across multiple domains.

- **Clinical Accuracy**: The model produces high-quality outputs.
  In hospitalization prognoses, it consistently distinguishes between cases requiring surgical intervention and those that do not, adapting recommendations accordingly. No clear hallucinations were observed, and the clinical justifications are coherent, concise, and well-aligned with the generated recommendations. While the model often omits to include secondary lifestyle advice (a tendency that mimics real life behaviors of physicians), it also avoids introducing irrelevant or low-value recommendations.

- **Instruction Adherence**: Instruction following is strong. The model reliably respects the required structure, and outputs are clear and interpretable for both medical staff and patients.

- **Conciseness**: The generated text is concise and stylistically aligned with clinical writing. It presents a mild tendency to overexplain in sections addressing hospitalization communication, though it maintains accuracy even in those cases.

- **Speed**: Latency is the primary drawback. Generation time ranges between 80 and 130 seconds per document, with noticeable variability depending on time of day in which it is used. This level of delay is difficult to reconcile with real-world clinical workflows, perhaps even moreso in its variation than in the time itself.

5.2.4 Claude Opus 4.6

A second high-capacity model accessed via the OpenRouter API, reporting state-of-the-art performance across a wide range of benchmarks. Unlike the other evaluated models, it is not free to use, with an approximate cost of 0.03$ per generated document.

- **Clinical Accuracy**: The model produces very high-quality outputs.
  In hospitalization prognoses, it consistently distinguishes between cases requiring surgical intervention and those that do not, adapting recommendations accordingly. No clear hallucinations were observed, and the clinical justifications are coherent, concise, and well-aligned with the generated recommendations.
  The model systematically includes secondary lifestyle recommendations (e.g. use of compression stockings in thrombotic conditions), which are often omitted in routine clinical practice due to time constraints. This represents a meaningful added value from the system.
  It does have a tendency to adopt a conservative approach, frequently recommending specialist consultations in potentially risky scenarios. While this is not clinically wrong behavior, it would not be sustainable in real-world healthcare settings due to healthcare system constraints, and would require manual filtering.
- **Instruction Adherence**: Instruction following is strong. The model reliably respects the required structure, and outputs are clear and interpretable for both medical staff and patients.
- **Conciseness**: The model exhibits a mild tendency to over-explain. Although this does not compromise clarity or correctness, it is a weak point that could use improvement.
- **Speed**: The average generation latency was approximately 16.5 seconds per document, making it the fastest model evaluated. Even accounting for review and correction, it would still constitute a significant improvement over manual drafting.

Summary The model produces near deployment-ready outputs, requiring only minimal post-editing. Despite minor tendencies toward conservatism and over-explanation, the results are extremely promising and further tweaking could render it production ready. Its primary limitation is cost.

5.2.5 Key Takeaways

The results suggest that, within this pipeline, output quality depends more on instruction-following ability than on domain-specific medical pretraining. MedGemma 4B-IT, despite being trained on medical data, produced the weakest outputs, largely due to its limited ability to adhere to structured formatting constraints at the 4B parameter scale but also due to severe hallucination problems.
In contrast, general-purpose models with strong instruction-following capabilities consistently generated more appropriate recommendations, even without explicit medical specialization. This is likely because the RAG setup shifts the burden of clinical knowledge from the model itself to the retrieved guideline excerpts, making generation primarily a task of organization and expression rather than clinical reasoning from parametric knowledge.

The results also point toward a clear performance gap between API-based and locally deployed models. However, this comparison emerges largely because only relatively small models could be tested locally due to hardware and budget constraints. Larger open-weight models are available and could potentially close this gap, while also addressing privacy concerns associated with API usage. This remains an important direction for future work.

Overall, while there is still room for improvement (discussed in the next section), the current results are encouraging and suggest that the system has realistic potential for deployment in clinical workflows.

6 Limitations and Potential Improvements

Several limitations emerged during development, mainly related to data quality, evaluation, and deployment constraints.

Guideline Quality One of the main bottlenecks comes from how clinical guidelines are structured. Key information, especially drug dosages and administration details, is often contained in tables or figures, which are not extracted in a readable manner by standard parsing tools such as PyMuPDF (Figure 5).

Table 12 Drugs for rate control in AF

Agent[a]	Intravenous administration	Usual range for oral maintenance dose	Contraindicated
Beta-blockers[b]			
Metoprolol tartrate	2.5–5 mg bolus over 2 mins; up to 15 mg maximal cumulative dose	25–100 mg twice daily	In case of asthma, non-selective beta-blockers should be avoided. Contraindicated in acute HF and history of severe bronchospasm.
Metoprolol XL (succinate)	N/A	50–200 mg once daily	
Bisoprolol	N/A	1.25–20 mg once daily	
Atenolol[c]	N/A	25–100 mg once daily	
Esmolol	500 μg/kg i.v. bolus over 1 min; followed by 50–300 μg/kg/min	N/A	
Landiolol	100 μg/kg i.v. bolus over 1 min; followed by 10–40 μg/kg/min	N/A	
Nebivolol	N/A	2.5–10 mg once daily	
Carvedilol	N/A	3.125–50 mg twice daily	

Figure 5: Examples of dosage recommendations in guidelines, these dosages are not repeated elsewhere in the document.

As a result, this information is lost during preprocessing, and the model is forced to produce incomplete recommendation. This is not a model issue, which was instructed to defer dosage to a specialist rather than produce one on its own, but a limitation of the parsing approach.
Future work should explore more robust solutions, such as LLM-based parsing or multimodal extraction.

Guidelines are also inherently not designed for emergency settings. They often include long-term treatments and diagnostic pathways (e.g. endoscopy results, oncology treatments) that are not relevant in acute care. While retrieval usually handles this reasonably well, some noise still makes its way into the final context.

Guideline Availability The guideline corpus is currently composed of 162 documents, but it is not complete. This is because clinical guidelines are published by different and independent medical societies with no standardization in publication methods, and are often difficult to access or collect systematically. In some cases, they are even inaccessible to the public and hidden behind a paywall. Despite efforts to gather a representative set, important gaps remain and are the direct cause behind poor retrieval performance in certain cases (e.g. pneumonia, malleolar fracture).
This is not a limitation of the system itself, but it does affect its observed performance. Medical expertise and guidance should be utilized to produce a complete informative medical corpus.

Evaluation Methodology The evaluation is limited both in size and approach. Unfortunately, only a very reduced number of clinical cases were available for use, which is not enough for a fully reliable assessment.

In addition, using LLM-as-a-judge for retrieval quality is an inconsistent method, with noticeable variability across and within models, beside their tendency to hallucinate. A more solid evaluation would require larger datasets.
Medical aid available in this project was unfortunately very limited due to their time constraints, and a decision to focus their clinical validation on the final output rather than on retrieval was made.

Budget and Hardware Constraints Hardware limitations had a significant impact on the experiments. Only small models could be tested locally, which made it necessary to rely on external APIs for stronger models. If shifted over to a real life scenario, this raises clear privacy concerns, as medical data, even when anonymized, remains of the most sensitive nature.
Running models locally would offer better control, but requires access to more powerful infrastructure. Infrastructure as a Service was considered as a potential alternative (ultimately negated due to both Microsoft Azure and Google Cloud Platform constraints on their student plans), but even then particular attention would need to be taken on the location of the physical server to ensure GDPR compliance.

System Scope At this stage, the project only covers the backend pipeline. No user interface has been developed, which obviously negates its usability in practice. A complete system would need a proper interface and integration into clinical workflows.
At the moment, the system only cites extracted excerpts with internal reference codes. Minimal adaptations could provide the user with links to the original document for immediate consultation.

Prompt Optimization Prompts were intentionally kept model-agnostic to ensure fair comparison. However, results show that models respond differently to the same instructions. Once a target model is selected, further improvements could likely be achieved through prompt tuning tailored to that specific model.

7 Conclusions

I believe this work shows that a RAG-based clinical text generation pipeline has the potential to aid Emergency Room physicians and patients alike in the production of more legible and comprehensive discharge recommendations, while also saving time for the medical staff. While the system could not exist as a standalone healthcare operator, and its outputs should always be reviewed by a medical expert, a conscious implementation could prove beneficial.

Results indicate that if high-quality clinical knowledge is supplied through retrieval, the generation task becomes primarily one of structuring, filtering, and expressing information rather than reasoning from internal knowledge. Modern Large Language Models are able to handle that task effectively, and relying on the provided information can significantly reduce the risk of hallucinations.
At the same time, the experiments make it clear that model performance alone is not the main bottleneck. The largest limitations arise from the quality and structure of the guideline corpus on which retrieval is performed. Missing information due to parsing failures, as well as complex guideline structure, directly impact the system's ability to produce correct and complete recommendations. Improving data extraction and curation is likely to yield larger gains than further marginal improvements in model capability.

From a practical standpoint, the system is already close to usable. High-end models are able to generate outputs that require only limited post-editing, and inference times, at least for some models, would result in an improvement compared to the current clinical workflow.

However, another constraint remains critical: data privacy. Reliance on external APIs is difficult to justify in real-world medical settings, making local deployment of sufficiently powerful models a necessary next step.

Were these improvements to be successfully implemented, a system of this kind could realistically move from experimental prototype to a practical clinical tool.

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A Example Appendix

A.1 Query Example

Included in this section is an example of how a set of queries is constructed from a clinical note for use in top-k retrieval and subsequent reranking with the Cross-Encoder. As previously stated, Query 3 is not used during the initial retrieval phase but only in the reranking stage, as empirical experiments showed this to be the best approach. The translated clinical note is also provided for reference.

All queries are available at the GitHub repository.

Queries

```
--- Query 1 (Diseases) ---
Diabetic fibrillation ATRIAL FIBRILLATION hypertension.
      ATRIAL FIBRILLATION OF NEW RAPID VENTRICOLAR RESPONSE.
      diagnosis comorbidities risk factors management

--- Query 2 (Medicinals) ---
Metoprolol tartrate amlodipine anticoagulant anticoagulant
      therapy metfromin therapy. ATRIAL FIBRILLATION OF NEW
      RAPID VENTRICOLAR RESPONSE. therapy recommendations
      pharmacological management dosage clinical scores
      diagnostic evaluation

--- Query 3 (Diagnostic Procedures - Condensed Clinical
      Diary) ---
CHA2DS2-VASc is calculated: 3 points (according to ESC 2020
      guidelines). Venous access found and troponin,
      biochemistry, blood count, TSH, pro-BNP, electrolytes
      performed. Pro-BNP 480 pg/mL (slightly elevated).
      ATRIAL FIBRILLATION OF NEW RAPID VENTRICOLAR RESPONSE.
      findings results clinical course management

--- Query 4 (Generic Query) ---
ATRIAL FIBRILLATION OF NEW RAPID VENTRICOLAR RESPONSE
      recommendations discharge management elderly male
```

Translated Clinical Note

```
MALE, 71 AA
Medical history:
Pathological proximate: Patient accesses for heart
      palpitation and sense of irregular heartbeat arising
      approximately 4 hours prior to access, associated with
      mild asthenia. Denies chest pain, syncope or pre-
      syncope. Reports similar, self-resolved episode about 3
       months ago, not investigated. Denies dyspnoea.
Remote pathology: arterial hypertension under therapy with
      amlodipine 5mg. Diabetic on metfromin therapy. No known
       dyslipidaemia. Former smoker, about 1 pack/day for 30
      years, STOP 15 years ago. No known cardiovascular
      events or family history of arrhythmias. Denies drug
      allergies.

Objective examination: Patient alert, oriented, cooperative.
       Mildly anxious. Eupnoic at rest. Vesicular murmur
      present in all lung areas, no added noise. Arrhythmic
      cardiac tones, pauses not free, heart rate approx.
      110-120 bpm. No appreciable murmurs. Abdomen treatable,
       not painful. Jordan and Blumberg negative. No
      declivous oedema or signs of deep vein thrombosis. BP
      158/94 mmHg. HR 114 bpm. SpO2 97\% aa. TC 36.5C.

Clinical diary:
- Immediate ECG is performed: rapid ventricular response
      atrial fibrillation (HR 114 bpm), no signs of acute
      ischaemia, no repolarisation changes. First documented
      finding.
- Venous access found and troponin, biochemistry, blood
      count, TSH, pro-BNP, electrolytes performed. Arterial
      EGA is performed: physiological exchanges, mild
      hypoxemia (PaO2 72 mmHg).
- Patient is sent to radiology for 2-projection chest X-ray:
       slight increase in hilar vascular pattern bilaterally,
       no frank pleural effusion, no parenchymal thickening.
      Mild enlargement of the cardiac shadow.
```

```
- Tests are viewed: troponin <14, negative. No significant
      indices of inflammation. Blood count normal. TSH normal
      . Pro-BNP 480 pg/mL (slightly elevated). Potassium 3.8
      mEq/L. Sodium 139 mEq/L.
- Bedside echo performed: no pericardial effusion. Modest
      left atrial dilatation. No diffuse B lines in lung
      fields. Visually preserved left ventricular systolic
      function.
- On-call cardiology is contacted for collegial assessment.
      Given the duration of the episode reported by the
      patient (approximately 4 hours), the approach is
      discussed: the patient is haemodynamically stable,
      therefore a decision is made to check the rate in the
      first instance.
- Metoprolol tartrate 5mg e.v. is administered under
      continuous monitoring. ECG check after 20 minutes:
      persistent AF, HR reduced to 88 bpm. Patient reports
      slight improvement in symptoms.
- Patient is informed of Diagnosis, need for anticoagulant
      therapy and related thromboembolic risk. CHA2DS2-VASc
      is calculated: 3 points (according to ESC 2020
      guidelines). Indication for oral anticoagulant therapy.
- Patient is admitted to cardiology for in-depth diagnosis,
      initiation of anticoagulant therapy and evaluation of
      rhythm vs. rate strategy in election.

Diagnosis: ATRIAL FIBRILLATION OF NEW RAPID VENTRICOLAR
      RESPONSE

Prognosis: hospitalisation
```

A.2 Prompt Examples

Included in this section are two example prompts generated by the IR pipeline: one for a hospitalization case and one for a home discharge. The differences in instruction design are voluntary and reflect the use of distinct templates created to adapt to different clinical scenarios.

These prompts represent the final input provided to the LLM for recommendation generation. For brevity, some parts of the examples have been condensed: the clinical notes are abridged, and only a single guideline excerpt is shown instead of the full set of ten retrieved passages. The complete prompts and outputs are available at the GitHub repository.

Hospitalization Prompt (Atrial Fibrillation)

```
ROLE:
You are an emergency department physician writing
      recommendations for a patient being admitted to
      hospital. The document will be read by the admitting
      team, the patient, and their family.

CLINICAL NOTE:

MALE, 71 AA
Medical history:
Pathological proximate: [...]
Remote pathology: [...]

Objective examination: [...]

Clinical diary:
[...]
- Metoprolol tartrate 5mg e.v. is administered under
      continuous monitoring. ECG check after 20 minutes:
      persistent AF, HR reduced to 88 bpm. Patient reports
      slight improvement in symptoms.
- Patient is informed of Diagnosis, need for anticoagulant
      therapy and related thromboembolic risk. CHA2DS2-VASc
      is calculated: 3 points (according to ESC 2020
      guidelines). Indication for oral anticoagulant therapy.
- Patient is admitted to cardiology for in-depth diagnosis,
      initiation of anticoagulant therapy and evaluation of
      rhythm vs. rate strategy in election.

Diagnosis: ATRIAL FIBRILLATION OF NEW RAPID VENTRICOLAR
      RESPONSE

Prognosis: hospitalisation


GUIDELINE EXCERPTS:

[E1] (AtrialFibrillation, p.20-20)

cardiac imaging, ambulatory ECG, other imaging as needed
      Assess new and existing risk factors and comorbidities
      (Class I) Stratify risk for stroke and thromboembolism
      (Class I) Check impact of AF symptoms before and after
      treatment (Class I) Assess and manage modifiable
      bleeding risk factors (Class I) Continue OAC despite
      rhythm control if risk of thromboembolism (Class I)
      Regular re-evaluation: 6 months after presentation, and
       then at least annually or based on clinical need
      Figure 3 Central illustration. Patient pathway for AF-
      CARE (see Figures 4, 5, 6, and 7 for the [R] pathways
      for first-diagnosed, paroxysmal, persistent and
      permanent AF). AF, atrial fibrillation; AF-CARE, atrial
       fibrillation-[C] Comorbidity and risk factor
      management, [A] Avoid stroke and thromboembolism, [R]
      Reduce symptoms by rate and rhythm control, [E]
      Evaluation and dynamic reassessment; CCS, chronic
      coronary syndrome; CHADS-VA, congestive heart failure,
      hypertension, age ≥75 years (2 points), diabetes
      mellitus, prior stroke/transient ischaemic attack/
      arterial thromboembolism (2 points), sodium-glucose
      cotransporter-2; VKA, vitamin K antagonist.

[E2] (AtrialFibrillation, p.24-25)
[...]

TASK:
The patient is being admitted to hospital. The ER does not
      prescribe ongoing therapy or follow-up, that is the
      admitting team's responsibility. Produce a brief
      handover note in two sections.

SECTION 1 - ADMISSION NOTE
In a numbered list:
- State that the patient is admitted to [department] for [
      key planned interventions], extracting the department
      and planned interventions from the clinical note (e.g.
      "Patient is admitted urgently to the Paediatrics
      department for high-dose intravenous immunoglobulin
      therapy, dedicated transthoracic echocardiography with
      coronary evaluation, and clinical-laboratory monitoring
      .").
- State that the patient and/or family have been informed of
       the diagnosis, the expected clinical course based on
      the guideline excerpts (e.g. treatment protocol,
      possible complications, need for monitoring), and the
      planned treatment pathway. Do not write examples.
- Do NOT give examples of complications or risks in
      parentheses. State the information directly without "e.
      g.", "such as", or "for example".
- If the patient is being hospitalised to perform surgery,
      state that the patient has been prepped.
- State that the patient is handed over to the care of the [
      department] team.
- Do NOT prescribe medications, follow-up visits, or
      lifestyle changes.
- Do NOT cite [E#] in this section.
- Keep each item to one or two sentences.

SECTION 2 - CLINICAL REFERENCES (for the reviewing physician
      )
For each item in Section 1 that draws on guideline content,
      write:
[number] - [E#] - brief clinical justification
Example:
2 - [E1][E9] - IVIG 2g/kg and aspirin 30-50mg/kg/day are
      first-line treatment for Kawasaki disease to prevent
      coronary artery aneurysms.


RULES:
- Use ONLY the provided guideline excerpts. Do not rely on
      prior medical knowledge.
- Do NOT invent clinical facts not in the clinical note or
      excerpts.
- Section 1 must be directly usable as a handover note with
      no editing needed.
```

Home Discharge Prompt (Thrombosis)

```
ROLE:
You are an emergency department physician writing discharge
      recommendations for a patient. The document will be
      read by the patient and their general practitioner.

CLINICAL NOTE:

FEMALE, 31 AA
Medical history:
- Pathological proximate: [...]
- Remote pathology: [...]
```

```
Objective examination: [...]

Clinical diary:
- Wells score for DVT is calculated: pain along deep venous
      course (+1), unilateral oedema (+1), swelling>3cm (+1).
       Total score: 3 - high pretest probability.
- Venous access found and D-dimer, biochemistry, blood count
      , coagulation performed.
[...]
- The angiologist/on-call internist is contacted for
      collegial assessment and therapeutic setting.
- Anticoagulant therapy is imposed: enoxaparin 8,000 IU (100
       IU/kg, weight 80kg) s.c., first dose administered in
      the PS.

Diagnosis: DEEP VENOUS THROMBOSIS OF THE LEFT GEMELLAR
      VENEES
Prognosis: 4 days


GUIDELINE EXCERPTS:

[E1] (VenousThrombosis, p.46-46)

which exhibits high risk clinical and/or anatomical features
      , a three month course of anticoagulation may be
      considered. Class Level Reference IIb C Nikolakopoulos
      et al. (2018) Recommendation 50 For patients with lower
       limb superficial vein thrombosis, acute superficial
      venous intervention is not recommended. Class Level
      Reference III C Lozano & Almazan (2003) Recommendation
      51 For patients with lower limb superficial vein
      thrombosis, ablation of incompetent superficial veins
      should be considered once the acute inflammatory and
      prothrombotic phase has settled, at least three months
      after the most recent thrombotic event. Class Level
      Reference IIa C Consensus 3. SPECIFIC TYPES OF VENOUS
      THROMBOSIS 3.. Upper extremity deep vein thrombosis
      3... Diagnosis. Approximately 10% of all cases of DVT
      occur in the upper extremities, which affect 4 e 10 per
       100 000 population. Two different types of UEDVT are
      recognised. Primary effort thrombosis (PageteSchroetter
       disease), often triggered by strenuous effort, and
      secondary UEDVT, which is usually related to the use of
       a central venous catheter (CVC), the latter being far
      more frequent (see Chapter 3.).


[E2] (VenousThrombosis, p.43-44)
[...]


TASK:
Produce discharge recommendations in two sections, do not
      add anything else.

SECTION 1 - DISCHARGE RECOMMENDATIONS
A numbered list of concise instructions for the patient.
Rules:
- One or two sentences per item. Be direct: "Continue...", "
      Stop...", "A prescription is issued for...".
- Combine closely related actions into a single item (e.g.
      stopping a medication and adopting an alternative is
      one recommendation, not two). Never combine medication
      prescriptions with non-medication actions.
- For medications: always include drug name, dose, route,
      and frequency (e.g. "every 12 hours"), but do not make
      up information. Copy the exact dosing from the clinical
       note. If frequency is not in the clinical note or
      excerpts, write "as directed by the specialist".
- For lifestyle factors (smoking, alcohol, diet, exercise):
      use strong advisory tone ("strongly recommended", "
      strongly advised") rather than direct orders.
- For lifestyle factors, if the patient is already doing the
       right thing (e.g. non-smoking, not overweight),don't
      acknowledge it. Only give instructions for things that
      need to be changed or started.
- For follow-up appointments and specialist referrals, use
      the phrasing: "A prescription is issued for [specialist
       type] evaluation for [clinical purpose]" and always
      specify the specialist type from the clinical note (e.g
      . "angiologist", "cardiologist", "internist") and the
      clinical purpose of the visit (e.g. "for thrombophilia
      screening", "for initiation of anticoagulant therapy",
      "for echocardiographic follow-up").
- Do NOT explain why. No rationale, no "because...", no "to
      reduce the risk of...", no pathophysiology.
- Do NOT speculate about future treatment decisions (
      duration of therapy, drug switches). Write "to be
      determined by the specialist" and move on.
- No sub-bullets, no bold, no markdown headers, no category
      groupings. Just a numbered list.
- Do NOT cite [E#] in this section. Citations belong only in
       Section 2.
- The second-to-last item must always be: "Go to the nearest
       emergency department immediately if you experience [
      relevant warning symptoms for this diagnosis]."
- The last item must always be: "You are referred back to
      your general practitioner for any further diagnostic or
       therapeutic needs and ongoing care."
- If a topic cannot be addressed by the excerpts: "Refer to
      specialist for [topic]."

SECTION 2 - CLINICAL REFERENCES (for the reviewing physician
      )
For each recommendation number in Section 1, write:
[number] - [E#] - brief clinical justification
Example:
1 - [E4][E7] - Anticoagulation recommended for confirmed
      distal DVT with high pretest probability.
3 - [E7] - Oral contraceptives classified as moderate VTE
      risk factor (OR 2-9).

RULES:
- Use ONLY the provided guideline excerpts. Do not rely on
      prior medical knowledge.
- Do NOT invent drug names, dosages, frequencies, or
      clinical facts not in the clinical note or excerpts.
- When multiple excerpts cover the same topic, write one
      recommendation and cite all relevant excerpts.
- Do not repeat the same recommendation in different wording
      .
- Section 1 must be directly usable as a discharge document
      with no editing needed.
```

A.3 Detailed Claude Outputs

Included in this section are two examples of outputs produced by the pipeline using Claude Opus 4.6 as the generation model: one for a hospitalization case and one for a home discharge. The outputs are presented in Italian, as the final system is designed to operate in that language and part of the evaluation concerns tone and clarity for patient-facing communication. The original clinical note, which was not accessible to the model at any stage of the pipeline, is also included for comparison. A brief commentary highlights the main strengths and weaknesses of the generated outputs.

All outputs from the various models are available in the GitHub repository, inside the `output [MODEL_NAME]` directories.

Hospitalization Output (Atrial Fibrillation)

```
## SEZIONE 1 - NOTA DI AMMISSIONE

1. Il paziente viene ricoverato nel reparto di Cardiologia
      per una valutazione diagnostica approfondita, l'inizio
      di una terapia anticoagulante orale basata su un
      punteggio CHAâĈĈDSâĈĈ-VASc di 3 punti e un processo
      decisionale condiviso riguardante il controllo del
      ritmo rispetto alla strategia di controllo della
      frequenza per una fibrillazione atriale di nuova
      insorgenza con risposta ventricolare rapida.

2. Il paziente e la famiglia sono stati informati della
      diagnosi di fibrillazione atriale di nuova insorgenza,
      del rischio tromboembolico associato che richiede una
      terapia anticoagulante, della necessitÃă di una
      gestione continua delle comorbilitÃă e dei fattori di
      rischio, compresa l'ottimizzazione del controllo dell'
      ipertensione e del diabete, e della valutazione
      pianificata delle opzioni di controllo del ritmo
      rispetto al controllo della frequenza durante il
      ricovero. Sono stati informati che sarÃă necessaria una
       rivalutazione regolare e che l'anticoagulazione orale
      deve essere continuata anche se viene ripristinato il
      ritmo sinusale, dato il profilo di rischio
      tromboembolico del paziente. Sono stati anche informati
       che il decorso clinico includerÃă ulteriori immagini
      cardiache, il monitoraggio ambulatoriale dell'ECG e la
      rivalutazione dinamica dei sintomi e della risposta al
      trattamento.

3. Il paziente viene affidato alle cure del team di
      cardiologia.
```

```
---

## SEZIONE 2 - RIFERIMENTI CLINICI (per il medico revisore)

1 - [E1][E4][E6] - Il paziente soddisfa i criteri per l'
      avvio di un anticoagulante orale con CHAâĈĈDSâĈĈ-VASc
      di 3 (ipertensione, diabete, etÃă âĽě65). Le linee
      guida raccomandano di stratificare il rischio di ictus
      e tromboembolismo (Classe I) e di iniziare l'OAC di
      conseguenza. Per la fibrillazione atriale persistente,
      si raccomanda un processo decisionale condiviso sul
      controllo del ritmo (Classe I), con un controllo della
      frequenza mirato alla frequenza cardiaca a riposo <110
      bpm come strategia alternativa. La cardioversione
      richiede un'adeguata valutazione dello stato di
      anticoagulazione, con OAC a breve termine per un minimo
       di 4 settimane dopo la cardioversione per tutti i
      pazienti.

2 - [E1][E2][E3][E8] - Le linee guida prevedono la
      valutazione e la gestione delle comorbilitÃă esistenti
      e dei fattori di rischio modificabili, tra cui l'
      ipertensione e il diabete mellito, che sono
      fondamentali per prevenire la recidiva e la
      progressione della FA. Il percorso AF-CARE richiede la
      gestione delle comorbiditÃă e dei fattori di rischio,
      la prevenzione di ictus e tromboembolie, la riduzione
      dei sintomi attraverso il controllo della frequenza e
      del ritmo e la valutazione con rivalutazione dinamica.
      L'imaging cardiaco e l'ECG ambulatoriale sono
      raccomandati come parte della valutazione iniziale. La
      continuazione dell'OAC nonostante il controllo del
      ritmo ÃÍ una raccomandazione di Classe I se il rischio
      tromboembolico persiste. Si raccomanda una
      rivalutazione regolare a 6 mesi dalla presentazione e
      successivamente almeno ogni anno. I fattori che
      favoriscono il controllo del ritmo includono il
      sospetto di tachicardiomiopatia, una breve anamnesi di
      FA e l'atrio sinistro non dilatato, tutti fattori che
      devono essere valutati durante il ricovero.

============================================================
ORIGINAL RECOMMENDATIONS (for comparison)
============================================================
Si consiglia:
- Paziente ricoverato presso U.O. Cardiologia per
      monitoraggio, ecocardiogramma
transtoracico dedicato, avvio di terapia anticoagulante
      orale (DOAC) e valutazione in
elezione della strategia terapeutica.
- Si informa il paziente e i familiari presenti della
      diagnosi e del percorso previsto.
```

Analysis While the model is clearly struggling with conciseness, it correctly identifies that no surgical intervention is required and avoids introducing inappropriate recommendations. Importantly, no hallucinated clinical information is observed, and all suggestions remain consistent with the provided context.

The verbosity is partly attributable to the use of model-agnostic prompting. Additional experiments showed that prompt adaptation specifically tailored to Claude significantly improved conciseness, at the cost of slightly degrading the performance of other models (the reason why those adapted prompts were not used for final evaluations). This suggests that further gains are easily achievable once a final model choice is made.

From a clinical perspective, conciseness remains secondary to clinical accuracy, which is the main strength of the model. Although there is room for improvement, the output maintains a high level of reliability.

It is worth noticing that hospitalization prompts are inherently less informative for evaluation purposes, as they follow a more rigid template. The key result is that the model correctly distinguishes cases requiring surgical intervention from those that do not, as confirmed by comparison with other diagnoses (e.g. appendicitis).

Home Discharge Output (Thrombosis)

```
## SEZIONE 1 - RACCOMANDAZIONI PER LA DIMISSIONE

1. Continuare l'enoxaparina 8.000 UI (100 UI per kg) per via
       sottocutanea ogni 12 ore; la durata totale della
      terapia anticoagulante deve essere determinata dallo
      specialista.
2. Interrompa la terapia contraccettiva orale con estrogeni
      e discuta con il suo medico di base o ginecologo di
      metodi contraccettivi alternativi non ormonali.
3. Si raccomanda vivamente di smettere di fumare
      immediatamente, in quanto il fumo ÃÍ un fattore di
      rischio significativo per la sua condizione.
4. Indossi una calza a compressione graduata sulla gamba
      sinistra, come indicato dallo specialista.
5. Si prescrive una valutazione da parte di un angiologo o
      di un internista per rivalutare la trombosi venosa
      profonda, definire la durata della terapia
      anticoagulante e fare uno screening della trombofilia.
6. Si prescrive una valutazione ematologica per lo screening
       della trombofilia, data l'assenza di un chiaro fattore
       provocante diverso dall'uso di estrogeni e dal fumo.
7. Monitorare i segni di trombocitopenia indotta dall'
      eparina: eseguire un emocromo con controllo delle
      piastrine entro 5 giorni dall'inizio dell'enoxaparina e
       successivamente secondo le indicazioni dello
      specialista.
8. Evitare l'immobilizzazione prolungata; si incoraggia una
      camminata leggera se tollerata, ma evitare un'attivitÃă
       fisica faticosa fino alla revisione dello specialista.
9. Si rechi immediatamente al pronto soccorso piÃź vicino se
       avverte improvvisa mancanza di respiro, dolore
      toracico, tosse con sangue, peggioramento del gonfiore
      o del dolore alle gambe, cambiamenti del colore della
      pelle dell'arto interessato o qualsiasi segno di
      sanguinamento insolito.
10. VerrÃă rinviato al suo medico di famiglia per qualsiasi
      ulteriore necessitÃă diagnostica o terapeutica e per l'
      assistenza continua.

## SEZIONE 2 - RIFERIMENTI CLINICI (per il medico revisore)

1 - [E3][E2] - L'anticoagulazione ÃÍ il trattamento standard
       per la trombosi venosa profonda del polpaccio
      confermata; la dose di enoxaparina ÃÍ coerente con il
      regime di eparina a basso peso molecolare basato sul
      peso, come da nota clinica. La TVP del polpaccio
      comporta il rischio di recidiva, soprattutto in
      presenza di fattori di rischio come l'eziologia non
      provocata.
2 - [E7] - La terapia contraccettiva orale ÃÍ classificata
      come un fattore di rischio moderato per il
      tromboembolismo venoso (odds ratio 2-9) e deve essere
      interrotta in caso di trombosi venosa profonda
      confermata.
3 - [E7] - Il fumo e l'uso di contraccettivi orali sono
      fattori predisponenti riconosciuti per il
      tromboembolismo venoso; la cessazione ÃÍ fortemente
      consigliata per ridurre il carico di rischio cumulativo
      .
4 - [E5] - La sindrome post-trombotica si sviluppa in una
      percentuale significativa di pazienti con trombosi
      venosa profonda degli arti inferiori; la terapia
      compressiva fa parte della gestione standard per
      attenuare la stasi venosa.
5 - [E2][E3] - La trombosi venosa profonda del polpaccio
      richiede un follow-up specialistico per valutare l'
      estensione prossimale, definire la durata dell'
      anticoagulazione e valutare i fattori di rischio di
      recidiva (etÃă, sesso, provocata rispetto a non
      provocata, coinvolgimento bilaterale rispetto a
      unilaterale).
6 - [E8] - Il test di trombofilia ÃÍ rilevante in questo
      giovane paziente con fattori provocatori limitati; le
      trombofilie ereditarie (fattore V Leiden, mutazione
      della protrombina G20210A, carenze di antitrombina/
      proteina C/proteina S) devono essere valutate per
      guidare le decisioni di anticoagulazione a lungo
      termine.
```

```
7 - [E4] - Il sistema di punteggio 4Ts guida la valutazione
      della trombocitopenia indotta dall'eparina; il
      monitoraggio delle piastrine durante la terapia con
      eparina ÃÍ una pratica standard, con una tempistica di
      caduta delle piastrine tipicamente tra il giorno 5 e il
       giorno 10.
8 - [E3][E2] - La mobilizzazione precoce ÃÍ generalmente
      raccomandata nella trombosi venosa profonda distale
      sotto anticoagulazione; l'immobilizzazione prolungata
      ÃÍ di per sÃĬ un fattore di rischio per la
      tromboembolia venosa.
9 - [E7][E2] - L'embolia polmonare ÃÍ una complicanza
      riconosciuta della trombosi venosa profonda; i sintomi
      di allarme (dispnea, dolore toracico, emottisi) e i
      segni di estensione del trombo o di complicazioni
      emorragiche giustificano una rivalutazione di emergenza
      .
10 - Raccomandazione generale per la continuitÃă delle cure.

============================================================
ORIGINAL RECOMMENDATIONS (for comparison)
============================================================
Si consiglia:
- Prosegua enoxaparina 8.000 UI s.c. ogni 12 ore fino a
      rivalutazione specialistica, oppure si
imposti terapia con DOAC secondo indicazione dello
      specialista.
- Sospenda immediatamente la pillola anticoncezionale e si
      rivolga al ginecologo per metodo
contraccettivo alternativo non ormonale.
- Si rilascia impegnativa per visita angiologica/ematologica
       e valutazione per trombofilia (da
eseguire a distanza dalla fase acuta e dalla sospensione
      dell'anticoagulante).
- Astensione dal fumo raccomandata con fermezza.
- Se comparsa di dispnea, dolore toracico, emottisi o
      peggioramento del quadro locale,
rientrare immediatamente presso il piÃź vicino PS.
- Si riaffida al medico curante per presa in carico e
      continuazione terapeutica.
```

Analysis Perhaps a more informative example is provided by the home discharge case, where the model is required to generate both therapeutic and lifestyle recommendations rather than simply deferring decisions to the admitting team.

The model correctly identifies the appropriate pharmacological therapy (enoxaparin 8000 UI every 12 hours). However, it is unable to specify the duration of treatment, which is a problem deriving from the previously discussed limitations of the guideline corpus. It also appropriately recommends discontinuation of oral hormonal therapy and suggests consultation with a gynecologist for alternative options.
Specialist follow-up visits are generally prescribed in a manner consistent with the original physician's recommendations, and accurately produces the right prescriptions. An exception comes from the model suggesting monitoring platelet count due to the risk associated with enoxaparin, since this prescription was not included in the original discharge note. However, the reviewing physician confirmed that while it reflects good clinical practice, it is not normally implemented in real-world cases to avoid overcharging the healthcare system. Although this would be a complicated issue to fix, such cases can be easily corrected during review by simply striking them out.

The model accurately identifies symptoms that should prompt re-access to the Emergency Room and expands on those listed in the original note without omitting any critical elements.
Lifestyle recommendations are perhaps the most interesting element, and represent a notable improvement: beyond maintaining advice on smoking cessation, the model introduces additional suggestions such as the use of compression stockings and light physical activity. According to the reviewing physician, these are often only communicated verbally (and sometimes omitted entirely) due to time constraints, and therefore their addition constitute a meaningful added value and potential contribution from the system.

Clinical references are clear and well synthesized. The system is designed to be able to provide direct links to the original guideline documents, rather than relying solely on reference identifiers (e.g., [E#]) as was done in the experiments, facilitating rapid verification by the reviewing physician.

While minor verbosity issues persist, they are less pronounced in this case. Further improvements are achievable through prompt optimization once a final model is selected.